Part 1: Team

Team Members and Assigned Roles

1) **Alexander Bondarenko** - Interviewer
2) **Maksim Beketov** - Observer
3) **Maksim Barannikov** - Note taker
4) **Varvara Orekhova** - Note taker
5) **Sofia Sokolova** - Note taker
6) **Pavel Dudinov** - Recorder

Summary of contributions:

1) **Sofia Sokolova**: organized group meeting, prepared questions for the interview, was involved in the discussion about the questions' restructuring, was a note taker during the interview, prepared a part of the product Research Board and Qualitative Analysis table.
2) **Maksim Beketov**: prepared questions for the interview, was involved in the discussion about the questions' restructuring, was an observer during the interview, prepared a part of the product Research Board and Qualitative Analysis table.
3) **Alexander Bondarenko**: was involved in the discussion about the questions' restructuring, was an interviewer, prepared a part of the product Research Board and Qualitative Analysis table.
4) **Maksim Barannikov**: prepared questions for the interview, was involved in the discussion about the questions' restructuring, was a note taker during the interview, prepared a part of the product Research Board and Qualitative Analysis table.
5) **Pavel Dudinov**: recorded the interview, prepared a part of the product Research Board and Qualitative Analysis table.
6) **Varvara Orekhova**: contacted the customer, prepared questions for the interview, was involved in the discussion about the questions' restructuring, prepared the script for the interview, was a note taker during the interview, prepared the transcript.

Part 2: Interview Script

Ask for permission to record the whole interview.
Let's start with your business goals.

1) For what purposes do you want to use this project?
2) Tell us about your experience with Face ID and other ways of face recognition. What are the pros and cons? What main characteristics from these examples do you want to see in your project, and what are strict nos? What features are not that important?
3) What main problem will be solved by moving from physical cards and passwords to such a system? What experience made you want to go to face recognition?
4) What is the threshold of success? What is the minimal set of requirements? What is the bonus task?
5) Failure of which parts of the project may lead to the failure of the full project?
6) Where exactly are you considering placing this face recognition camera: in an office, or in a house? For what purposes will your clients use this project?
7) Why is your project unique, and why does it have to attract clients?

Let's continue and talk about project scope.

8) For this project, do you want different types of users? If yes, what exact types will there be?
9) What do you want these users to be able to do (depending on their status)?
10) What are the requirements for the database: how many faces does the system have to be able to store?

Now, we will discuss features.

11) What is the threshold of success for the working system? For example, we are testing a system, how many false rejects per N attempts is acceptable. What is expected for the minimal level of precision/recall?
12) What is more important to you? If the system makes the recognition fast, but may mistake the wrong person for the right one, or full security, where there are no false acceptances but there is a possibility that the registered person will be rejected?
13) How do you expect the registration of a new face to go? Who has to be able to do it? Can you describe step-by-step how this registration has to go?
14) Is there good lighting in a place where you want to put the camera? How is the light in the morning and in the dark of night?

15) What is the user's behaviour while recognizing, does it have to be in motion, or while standing in one place?
16) What is the maximum delay in the system's deciding whether it's the "known/unknown" person, and what time for turning on the face recognition when the face gets in the camera?
17) How many sides of the face are required to be saved for creating a secure base?

The next topic is the implementation.

18) What camera and servo will be given?
19) What is the way to access the camera on Raspberry Pi 5 do we have to use?
20) Do we need to store photos of faces, or just embeddings?
21) How does the laptop version have to work without GPIO?
22) How will the Docker multi-arch be checked?
23) What exactly has to be written in the final report and README?

If we have time, we would like to ask additional questions.

24) Your requirements mention protection from photos. Is it acceptable in case of the system's doubt to ask a person to smile, or make a serious face, or make some gesture?
25) If a person was rejected, is it needed to send a report about this case to the admin?
26) Does the system need to display the user's name or confidence level?
27) Do you need an option for giving temporary access to a person, for example, for 24 hours?
28) What should be the percentages of successful system triggers, if the user is wearing some attributes, like glasses, masks, and hats?
29) Is liveness detection necessary?

Lastly, let's talk about constraints:

1) What are the hard deadlines?
2) What tech stack must be used?

Part 3: Mom Test Principles

Our team began with an initial list of questions created by individual team members. Then, we combined these lists, removed duplicates, and reviewed each question against Mom Test principles. Using the Mom Test, we tried to avoid asking about future plans or opinions, and ask more about past experiences, their life, and present problems. Example of questions that were improved:

before Mom Test	after Mom Test
What is the objective of your company?	For what purposes do you want to use this project?
Why do you want to build this product?	What experience made you want to switch to face recognition?
What do you want users to be able to do?	Tell us about your experience with Face ID and other ways of face recognition. What are the pros and cons?
What is the threshold of success?	For example, we are testing a system. How many false rejects per N attempts is acceptable? What is expected for the minimal level of precision/recall?
What are the existing challenges of such a product?	What annoys you the most about the current system?

Part 4: Transcript

I - Interviewer
C - Customer

I: Thank you for your time. May we record this interview?
C: Yes, no problem.

I: Let's start. For what purposes do you want to use this project?
C: To make it easier to give people access to the basement and the laboratory. Currently, people use cards, but it takes a lot of time to grant access to a new person. So, the location is the university basement, specifically laboratories.
I: So, in the whole university?
C: Yes, but for now, only to test here.

I: Tell us about your experience with Face ID and other ways of face recognition. What are the pros and cons? What main characteristics from these examples do you want to see in your project, and what are strict nos? What features are not that important?
C: Honestly, I have not worked with any such framework. I only know that there is a working model.

I: What main problem will be solved by moving from physical cards and passwords to such a system? What experience made you want to go to face recognition?
C: It will solve the problem of the time it takes to grant access. It will be faster than making cards and waiting for them.

I: What is the threshold of success? What is the minimal set of requirements? What is the bonus task?
C: The basic requirements are: capture face, store it in the database, check when a person enters. The bonus task is liveness check. The system has to reject photos. At the same time, the system has to accept the right people with glasses, masks, and makeup.

I: Failure of which parts of the project may lead to the failure of the full project?
C: I see none.

I: Where exactly are you considering placing this face recognition camera: in an office, or in a house? For what purposes will your clients use this project?
C: Only for the university.

I: Why is your project unique, and why does it have to attract clients?
C: It doesn't have to. It is a functional system, not a product.

I: What exact types of users will there be?
C: There is an admin who has rights to access the box via SSH key, add a new user, delete a user. And there are other users who just scan their faces.
I: So, the hierarchy: admin and users?
C: The one that has access to Raspberry Pi has access to users.
I: Got it.

I: Now, we will talk about features. What are the requirements for the database: how many faces does the system have to be able to store?
C: Usually, we have about 30 people here.
I: Okay, but about the university, we have about six thousand people here. What about that?
C: Still, not so many people go down here.
I: Okay, got it.

I: What is more important to you? If the system makes the recognition fast, but may mistake the wrong person for the right one, or full security, where there are no false acceptances but there is a possibility that the registered person will be rejected?
C: Definitely security.

I: How do you expect the registration of a new face to go? As you said, only the admin can do this, correct? Can you describe step-by-step how this registration has to go? Will it be the same example with the box?
C: Only the admin can add a new user. The algorithm is to get connected to the box, the person is in front of the camera, the data from the camera is extracted and added to the database. Then, the person is granted access.

I: Okay, then what about the user's behaviour while recognizing, does it have to be in motion, or while standing in one place?
C: Do what is more comfortable. You can do it in motion, but it may be harder.
I: I agree with you. And as I can see, the doors here are not opening automatically, so it is better to check people while standing.

I: What is the maximum delay in the system's deciding whether it's the "known/unknown" person, and what time for turning on the face recognition when the face gets in the camera?
C: 30 milliseconds to process the face and video frame. The final response is 30-40 milliseconds. 150-200 milliseconds is for model processing. Overall, 1-2 seconds is acceptable. There is no strong bottleneck.
I: So, it is not a strict requirement?
C: But I don't want to wait for two minutes.

I: How many sides of the face are required to be saved for creating a secure base?
C: You need to find this out experimentally.

I: Okay, then the next question, what camera and servo will be given?
C: Servo is blue SG 90. The camera is a standard webcam.

I: Do we need to store photos of faces, or just embeddings?
C: Only embeddings.

I: How does the laptop version have to work without GPIO?
C: A development build to check hypotheses and run tests without hardware.

I: How will the Docker multi-arch be checked?
C: It just needs to work on both x86 and ARM.

I: Another question, what exactly has to be written in the final report and README?
C: The README should contain a setup guide. The final report is a demonstration that the project works. It should show all working features.

I: What should be the percentages of successful system triggers, if the user is wearing some attributes, like glasses, masks, and hats?
C: The system should have 95% or higher successful recognition rate on average, and in case with coverage it has to be 60-70%

I: Do we still have time? Can we ask you some additional questions?
C: Yes, no problem.

I: Your requirements mention protection from photos. Is it acceptable in case of the system's doubt to ask a person to smile, or make a serious face, or make some gesture?
C: Usually we use pupil movement for liveness detection, but you can also do it with a smile check.

I: If a person was rejected, is it needed to send a report about this case to the admin?
C: It would be good to log it.

I: Does the system need to display the user's name or confidence level?
C: Only in the logs.

I: Do you need an option for giving temporary access to a person, for example, for 24 hours?
C: Yes, it is a useful function, and I want it.

Then, the Interviewer finished and Observer and Note takers asked some additional questions that were missed.

I: Question is about lightning. Is there good lighting in a place where you want to put the camera? Will it be turned off at night here?
C: The lighting is usually static. You can rely on that level of lighting. But the system still needs to be robust. It should work both with warm and cold lighting.

I: Does the camera have a flash?
C: No.

I: Will you give us the database to train the model? Or, we need to find it ourselves?
C: You can do it by yourself.
I: Can we use our faces?
C: Yes.

C: Actually, about the flash, the idea is not bad. There is no flash in the camera, but you can make it yourselves and use it.
I: Okay, we will think about it.
I: Do we need to have a message about turning on the flash?
C: I don't have such a requirement.

I: What operating system should be installed on Raspberry Pi?
C: Any that you want.

I: How much memory will the SD card in the camera have?
C: 32 or 64 GB.

I: Also, can we clarify about the work. Will you help us?
C: Yes, if you have some problems with hardware and need hints, you can ask me anytime, just write to me. Regarding the hardware: when you are ready, just write to me, and I will give it to you. You cannot take it out of this laboratory.

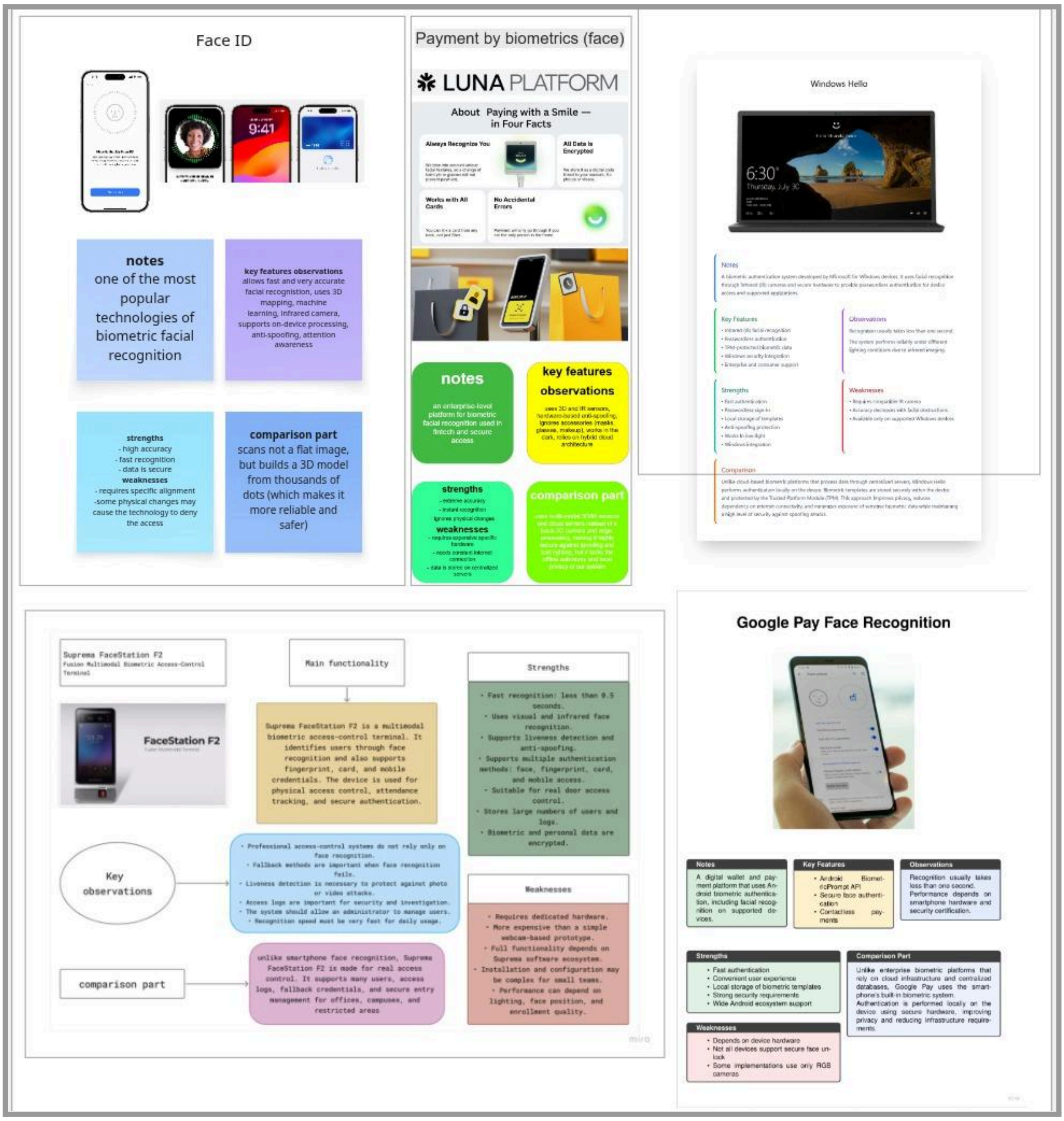
I: That's all. Thank you very much!
C: Thank you.

Part 5: Interview Recording

[link]

Part 6: Product Research Board

[link]



Part 7: Qualitative Analysis Table

[link] - for better view

feature/product	**Face ID**	**Payment by biometrics (face) (LUNA Platform)**	**Windows Hello Face Recognition**	**Google Pay (Face Recognition)**	**Suprema FaceStation F2**
base technology	TrueDepth camera and machine learning	Hybrid cloud. Custom hardware from the terminal and integration with 2D RGB and 3D/IR cameras (LUNA also integrates with any IP-cameras)	Infrared (IR) camera, RGB camera, depth sensing (depending on device), and machine learning. Uses biometric authentication integrated into Windows.	Uses Android BiometricPrompt API and device-integrated facial recognition systems. Depending on the device, recognition may rely on RGB cameras, infrared sensors, depth mapping, and machine learning algorithms.	Fusion multimodal access-control terminal. Uses visual face recognition and infrared (IR) face recognition with a deep learning algorithm. Supports several credentials: face, fingerprint, cards, NFC/BLE mobile access, and Template on Card
failure threshold	5 attempts before a passcode is required	The False Acceptance Rate approaches zero (On LUNA, it can be adjusted flexibly depending on the task.). In the case of Sberbank, if the system doubts the authenticity of the data, it asks for the "smile code" that the user created	After several unsuccessful attempts, Windows requires a PIN, password, or another configured authentication method.	The number of failed attempts depends on the device manufacturer. After multiple unsuccessful attempts, the user must enter a PIN, password, or pattern.	Authentication policy can be configured by the administrator. If face recognition fails, the user can use alternative credentials such as fingerprint, access card, or mobile credential depending on

					the selected model and configuration
recognition with accessories	Recognizes	Recognizes	Recognizes	Recognizes	Recognizes
distance and angle	25–50 cm, straight-on view (up to 20–30 degrees), eye-level angle. Extreme angles or flat surfaces can cause recognition failures. Also, the camera needs to clearly see both your eyes, nose, and mouth.	30-50 cm, the terminal's camera has a fixed tilt level and can handle head rotations of up to 30-45 degrees. (LUNA recognizes faces in dense traffic)	Typically 30–80 cm from the camera. Works with moderate head rotations and different lighting conditions thanks to IR sensors.	Typically works at 25–60 cm from the face and tolerates moderate head rotations. Exact limits depend on the smartphone model.	Recognition distance is 50–130 cm. The device supports walkthrough facial authentication, so the user does not need to precisely fit the face into a fixed area
expected time for recognition	About 0.5 - 0.8 second	About 1 second	About 1 second or less	Less than 1 second.	Less than 0.5 seconds
model improvement	With each successful unlock, it updates the mathematical representation of the face to adapt to the changes. Face ID will also update this data when it detects a close match but a passcode is subsequently entered to unlock the	Centralized updates. The vendor collects data and deploys new model versions to the servers.	The biometric template adapts over time through successful authentications and can learn appearance changes such as new glasses or facial hair	Facial recognition models are continuously improved through Android security and machine-learning updates provided by Google and device manufacturers.	Uses a deep learning-based face recognition algorithm. Recognition quality can be improved through better enrollment, firmware/software updates, threshold configuration, and stable installation conditions

	device.				
data protection	Face ID data (including mathematical representations of the face) is encrypted and protected by the Secure Enclave	Biometrics in the form of encrypted vectors are stored in the Unified Biometric System (UBS) and on the bank's secure servers.	Biometric templates are stored locally on the device, protected by the TPM (Trusted Platform Module) and Windows security architecture. Facial images are not stored directly.	Biometric data is stored securely on the device and protected by Android's hardware-backed security mechanisms. Google Wallet requires strong (Class 3) biometric authentication for payment authorization.	Biometric credentials and personal information are encrypted. The device also uses secure boot protection
probability of misrecognition	Less than 1 in 1,000,000	Extremely low (no precise data, tends to zero)	Very low. Microsoft states that Windows Hello provides enterprise-grade biometric security and significantly reduces unauthorized access compared to passwords.	Extremely low for devices certified with Android	Very low. Suprema states a False Acceptance Rate (FAR) of 1 in 10 billion for FaceStation F2
liveness recognition	The projector emits 30,000 invisible dots to accurately determine the geometry and volume of the face. The system cuts off a flat photo or mask immediately. The camera tracks blinking, eye movement,	To recognize a person, a mathematical model is used, which is created based on a photograph. Moreover, built-in Time-of-Flight or infrared sensors read depth (3D map) by physically blocking flat photo screens. (In the case of the LUNA, 2D frame analysis	Uses infrared imaging, depth information, and anti-spoofing mechanisms to distinguish a real face from photographs or videos.	Advanced implementations use depth information, infrared sensors, or AI-based anti-spoofing techniques to distinguish a real person from photos or videos.	Supported. Uses deep learning-based live face/fingerprint detection and anti-spoofing technology

	and micro-facial changes in real time. Unlocking requires a person to consciously look at the screen (this setting can be disabled in the Universal access settings, but it is active by default for security)	algorithms can also be used.)			
field of usage	Quick unlocking of devices (iPhone, iPad), confirmation of online and offline payments, authorization in applications (banking, messengers), and auto-completion of passwords	Safe city, finance (fintech, retail, cash desks), transport, industry	Windows device sign-in, enterprise authentication, passwordless login, access to applications and corporate resources.	Contactless payments, online purchases, app authentication, account verification, and authorization of financial transactions	Enterprise physical access control, office entrances, campuses, restricted rooms, attendance systems, security checkpoints, and contactless authentication scenarios
lighting	Works in the dark, but extremely harsh backlighting or glare can sometimes disrupt the infrared sensors	The terminal's infrared illumination allows for confident facial recognition in any lighting conditions. (In the case of the LUNA, it depends on the camera's features)	Works in both bright and low-light environments due to the use of infrared (IR) cameras. Performance is generally unaffected by ambient lighting conditions.	Performance depends on device hardware; infrared-equipped systems can operate in low-light conditions.	Works in indoor lighting conditions. IR recognition allows authentication under dim lighting. Performance is better in stable indoor lighting and may decrease

					in extreme lighting conditions
additional comments	The most popular and safest technology, but it lacks long-distance (1m) recognition	Expensive infrastructure. Closed source code requiring bank-grade security certification.	Requires certified Windows Hello hardware. Closed-source implementation integrated into the Windows security ecosystem.	Closed source code requiring bank-grade security certification.	Very relevant to FaceGuard because it combines face recognition, access decision, multiple authentication methods, anti-spoofing, user database, logs, and physical access-control integration
type of camera	TrueDepth system (3D structured light and infrared camera)	Terminal: 2D RGB and 3D/IR cameras. LUNA: any IP-cameras and 3D/IR modules	Infrared (IR) camera with RGB camera support and optional depth-sensing hardware.	Depends on the smartphone: RGB camera, infrared camera, depth sensor, or a combination of these technologies.	Built-in visual and infrared face recognition camera system
data storage	Calculations on a local coprocessor, the mathematical template is encrypted and lies locally	Hybrid or server architecture is used (Hybrid Cloud). The data is centralized in the Unified Biometric System	Biometric templates are stored locally in encrypted form and protected by TPM hardware. Authentication is performed on-device.	Biometric templates remain on the user's device and are not uploaded to Google Pay servers. Authentication is performed locally by the operating system.	Supports up to 100,000 users, 50,000 users for facial authentication, 50,000 image logs, and 5,000,000 text logs

Part 8: Questions for Clarification

1) At what height will the camera be installed, at the eye-level or higher?
2) Should the administrators be able to change the user's role? For example, promote someone to admin, or demote them.
3) Should access outcomes trigger any notification, sound, or color indicators?
4) What should happen if many faces appear in the camera: reject all, choose the closest, or wait?
5) How long should the system continue working during a power outage?
6) Should the system store login attempts locally when there is no internet connection, such as successful access, denied access, recognition errors, or camera failures, and synchronize these records after the connection is restored?
7) What should the system do if the camera is unavailable or stops sending frames?

Part 9: Core Features for MVP

1) **Face Adding**. An admin can register a new face in the system by capturing the new face on camera and saving its embedding.
2) **Face Recognition and Access Decision**. The system can analyze the face and verify whether it is in the database.
3) **User Deletion**. An admin can remove a person from the database.
4) **Access Logging**. All access attempts are recorded in the logs.
5) **Development Build**. This build works without physical hardware and allows the system to be tested and debugged without Raspberry Pi.

Part 10: Use of AI/LLM

AI was used to generate visual style for the two research boards out of the total five. In all other parts of the assignment no AI was used.